Cisco Packet Tracer

Network Infrastructure Project Report

Screenshot-by-screenshot explanation of network design, addressing, DHCP, HTTP and connectivity testing

Environment	Cisco Packet Tracer simulation
Departments	IT, Sales and HR
Devices shown	1 router, 3 switches, 6 PCs and 1 server
Evidence covered	Topology, IP configuration, DHCP, HTTP and ping tests

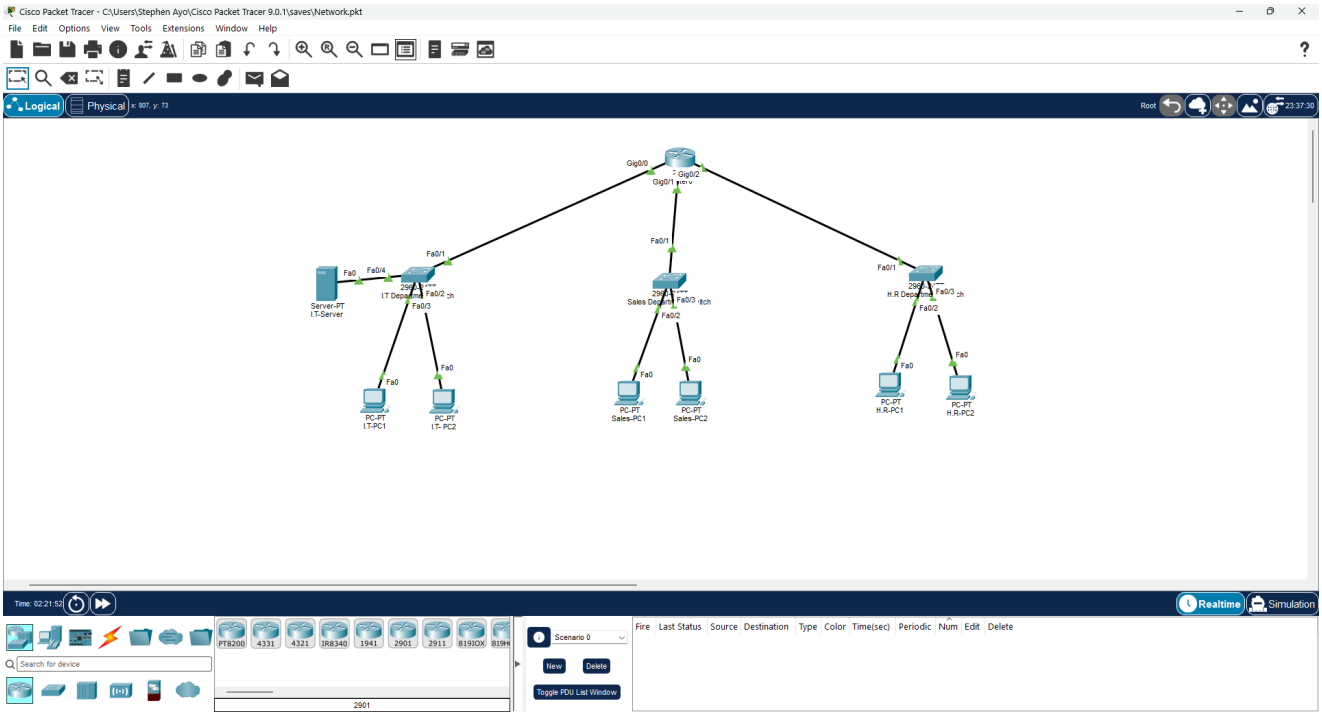
Purpose of this report

This report explains what the supplied screenshots demonstrate and how each relates to the project. Failed DHCP requests and ping attempts are documented as troubleshooting evidence rather than presented as successful tests.

The report does not claim that FTP was successfully tested or that the full ACL configuration was independently verified. Those items require separate evidence if they are part of the assignment requirements.

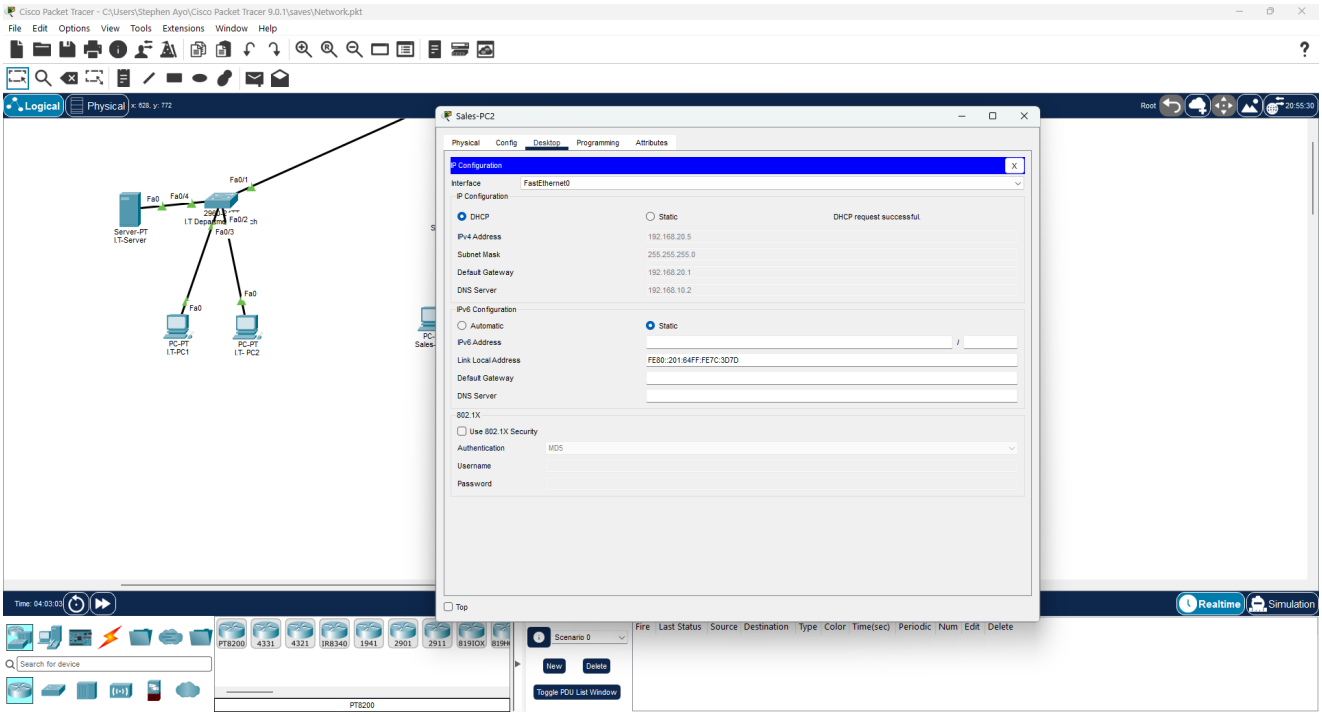
Screenshot 1 — Overall network topology

The overall design shows one router connected to three departmental switches for IT, Sales and HR. Each department has two PCs, and the IT network also contains a server. This demonstrates the planned segmented network layout and links between devices.



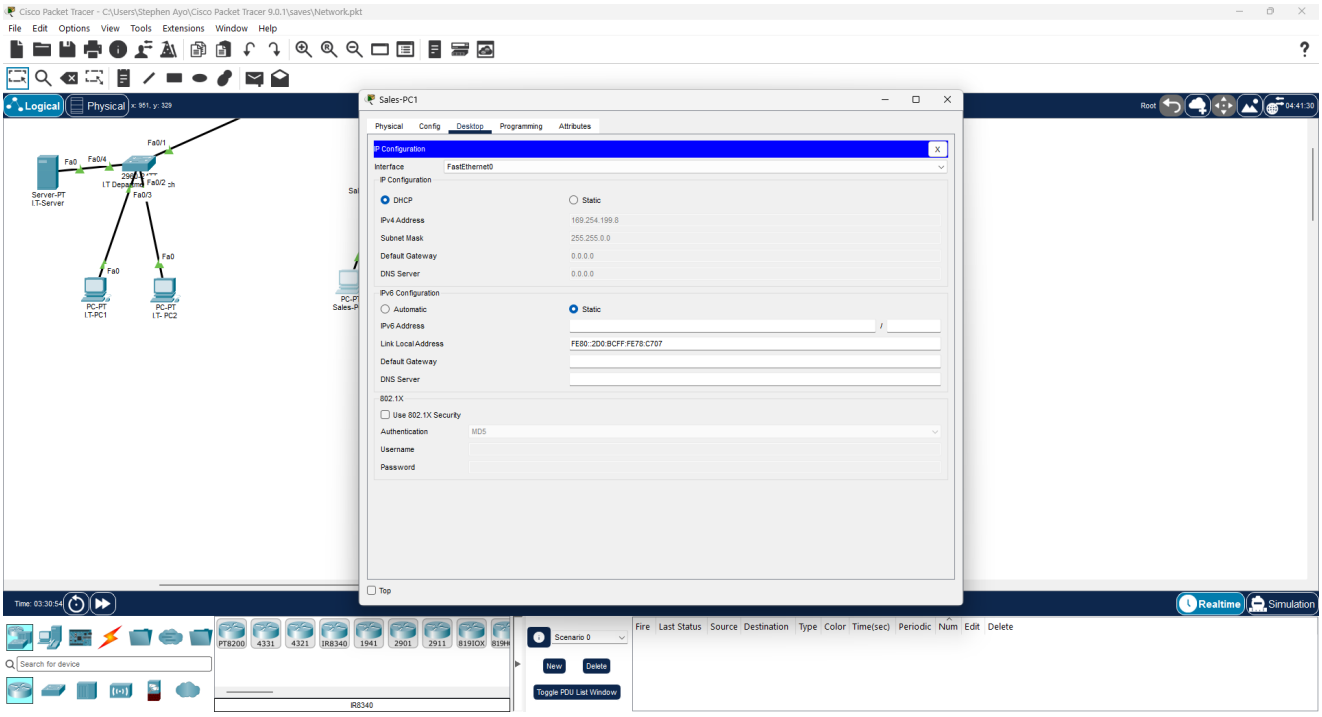
Screenshot 3 — IT server static IP configuration

The server is configured with IPv4 address 192.168.10.2, subnet mask 255.255.255.0 and default gateway 192.168.10.1. A fixed address gives clients a predictable destination for server services.



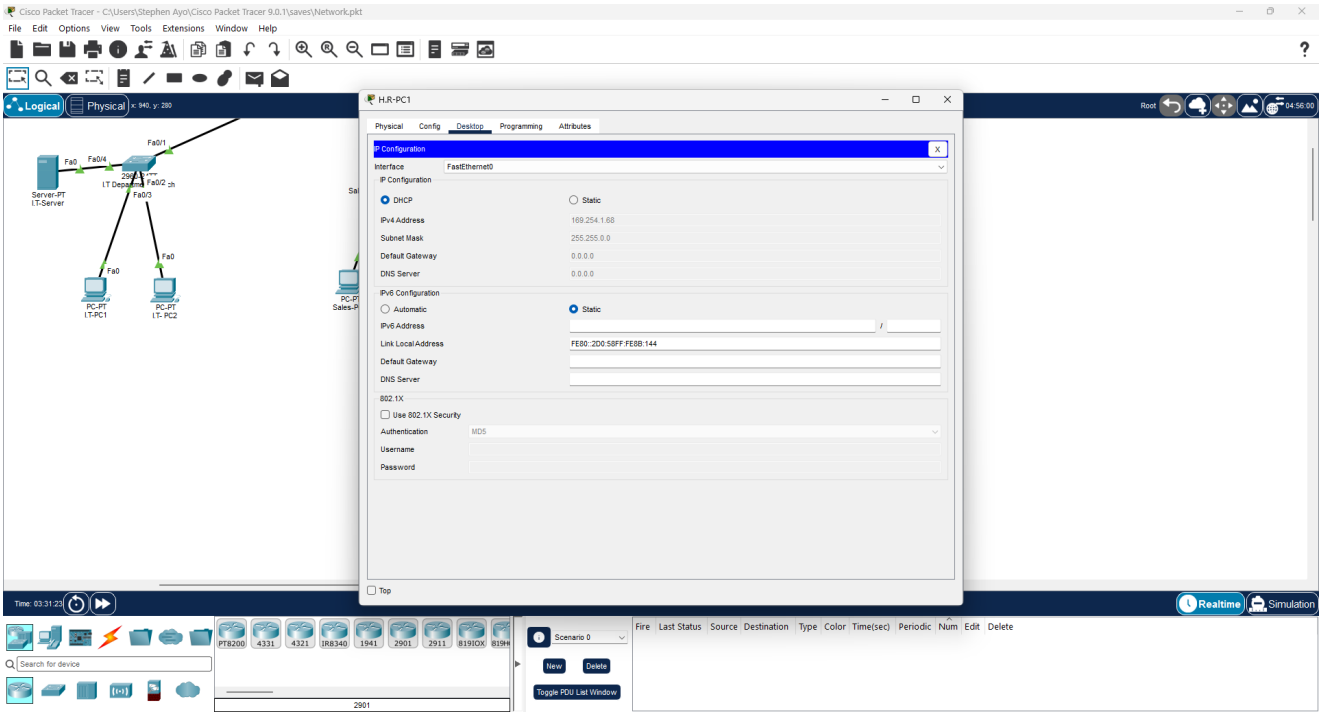
Screenshot 4 — Client IP configuration check

This is a client IP-configuration screen from the addressing checks. DHCP is selected so the client can request network settings automatically. Treat it as evidence of the configuration screen, and rely on captures showing an assigned address for proof of a successful lease.



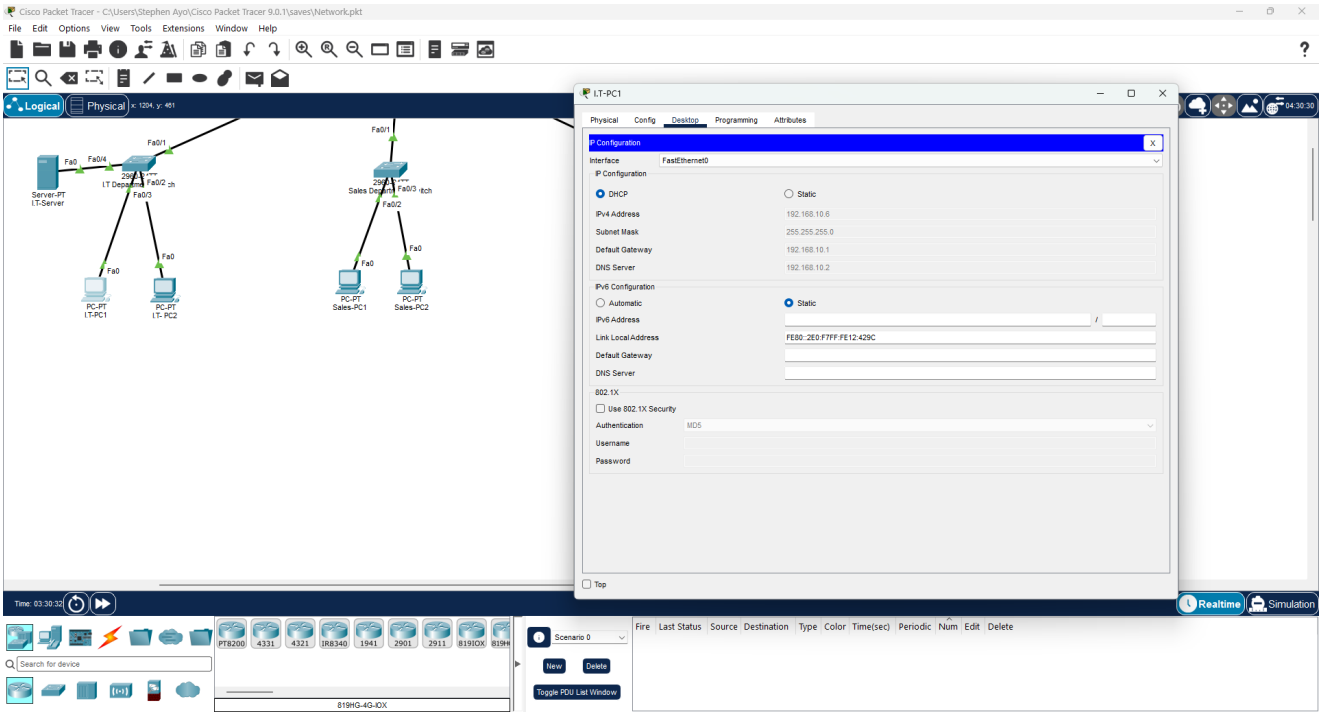
Screenshot 5 — Additional client IP configuration

This additional client configuration capture documents the addressing checks across the network. Similar captures may show different stages, so successful lease screenshots are the strongest evidence that DHCP worked.



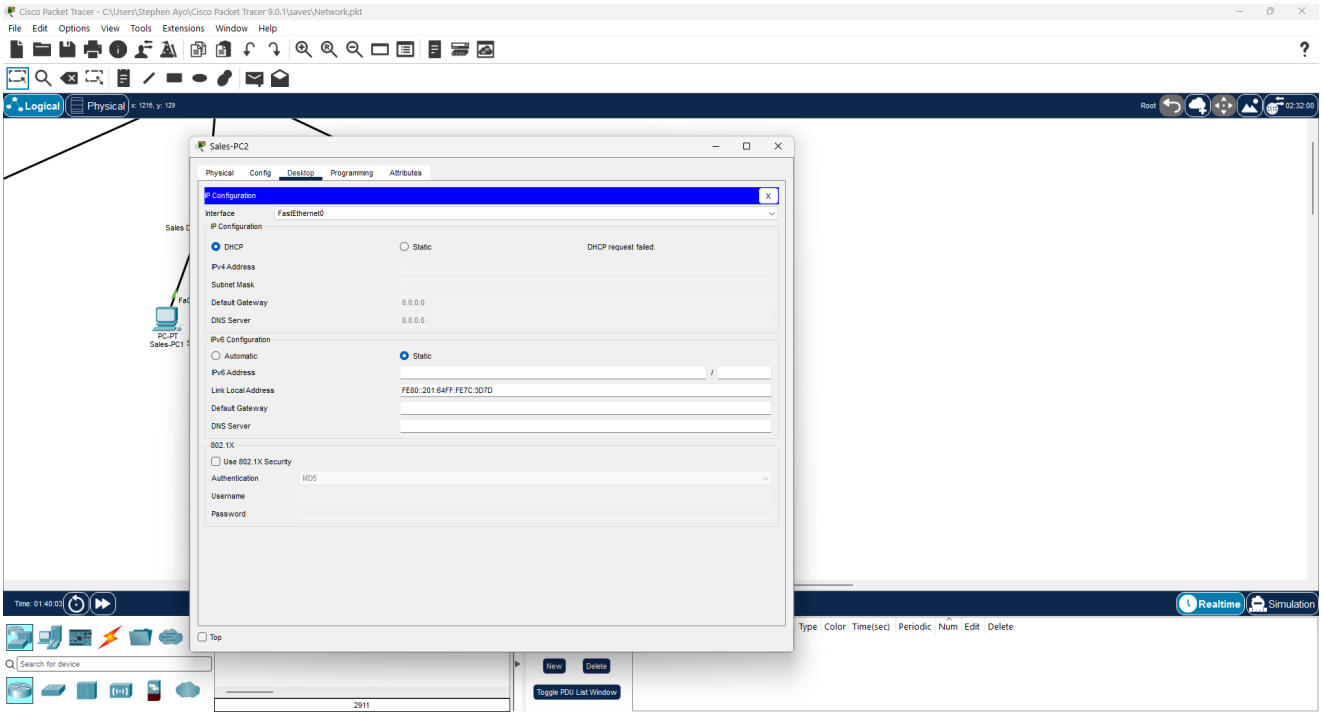
Screenshot 6 — DHCP lease on an IT PC

The IT PC is shown using DHCP with IPv4 address 192.168.10.6, subnet mask 255.255.255.0, default gateway 192.168.10.1 and DNS server 192.168.10.2. This is evidence of a successful DHCP lease for an IT client.



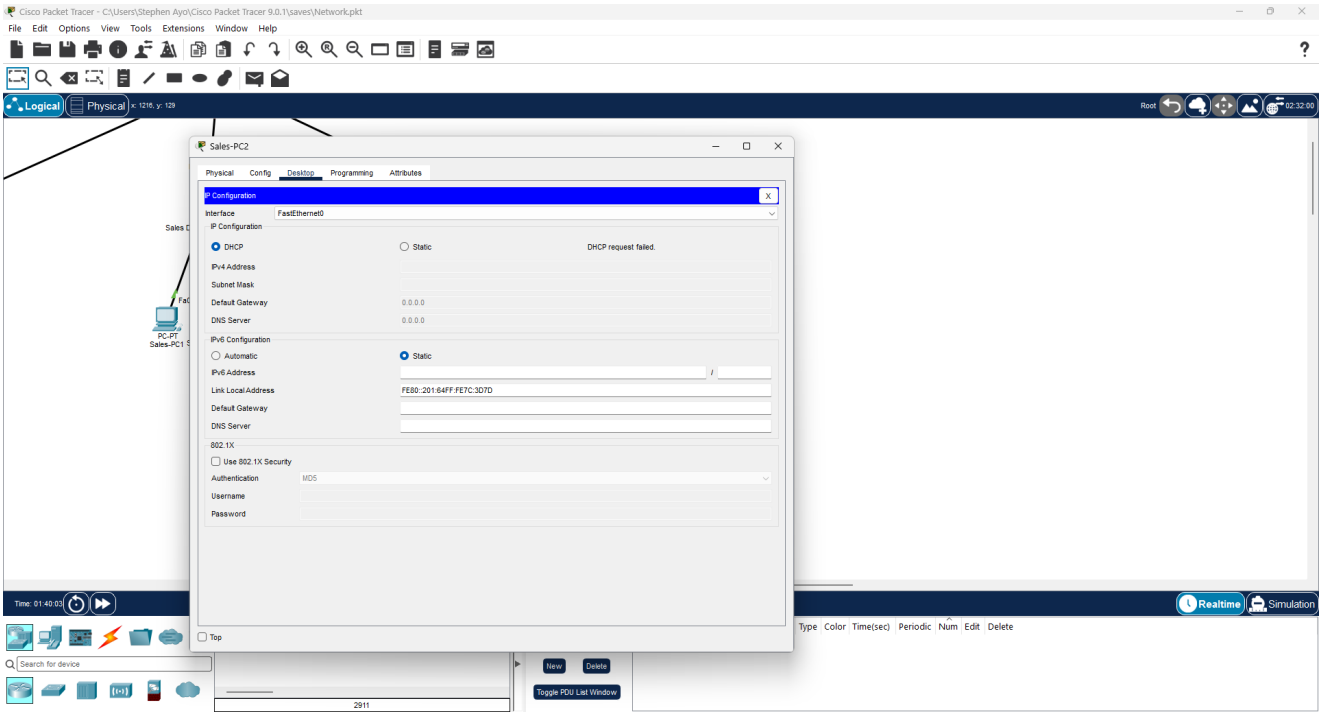
Screenshot 7 — DHCP request failure

This screen shows a DHCP request failure. It records an earlier troubleshooting stage when the client did not receive an IPv4 lease. Keep it as evidence of the problem encountered, not as proof of successful DHCP.



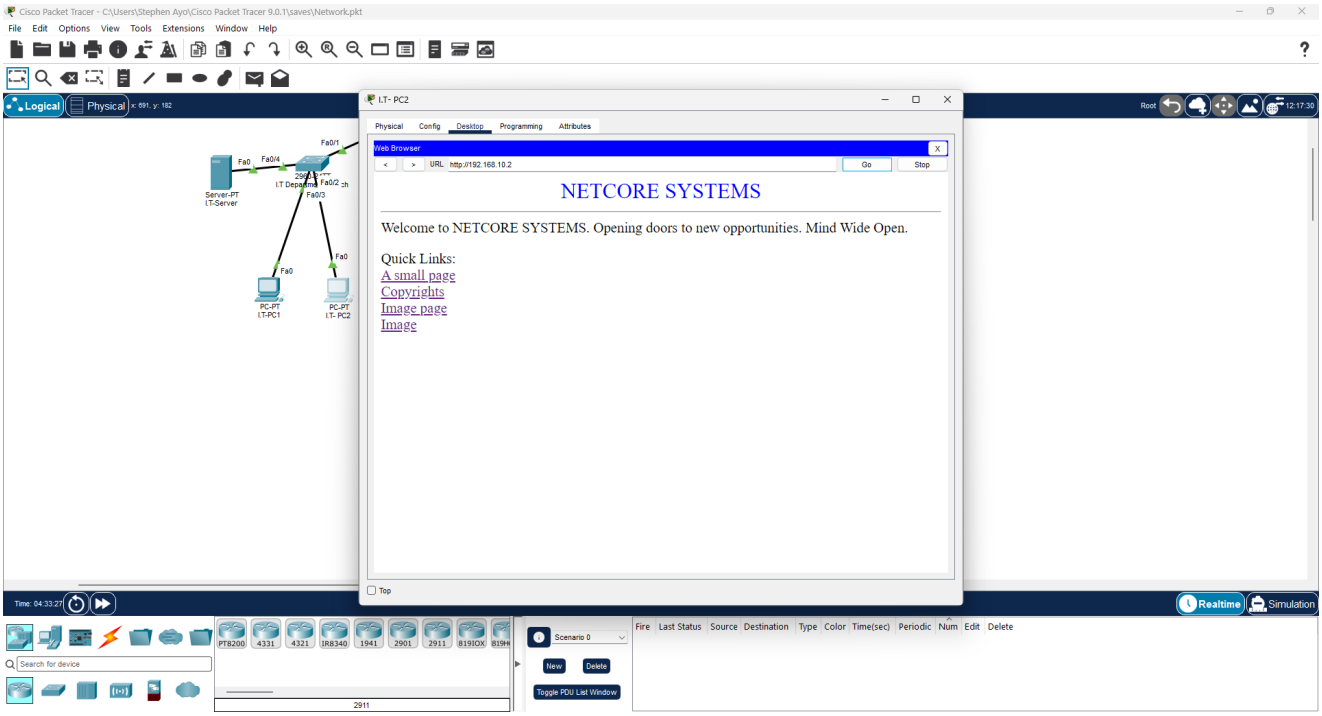
Screenshot 8 — Repeated DHCP troubleshooting capture

This is a further capture from the DHCP troubleshooting stage. It documents the earlier issue before a working lease was obtained and should be labelled as an intermediate troubleshooting screenshot.



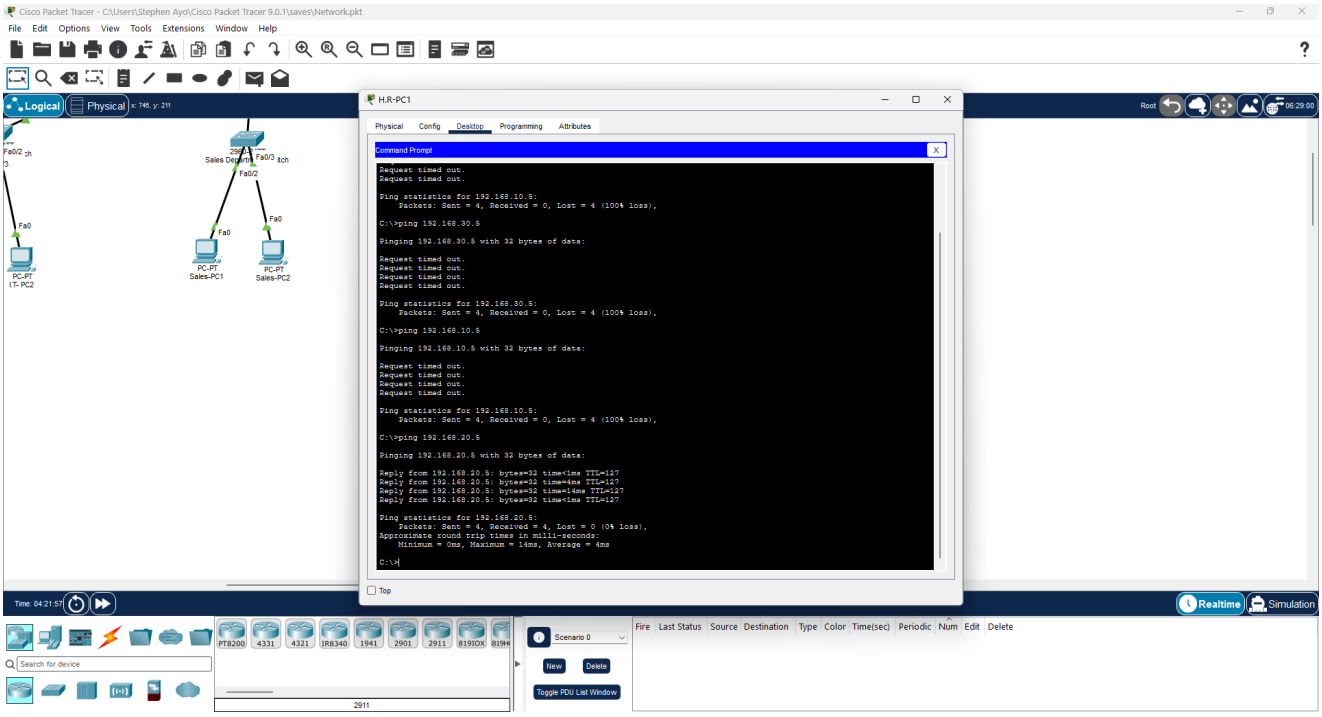
Screenshot 10 — HTTP website test

The browser is pointed at http://192.168.10.2 and displays the NETCORE SYSTEMS website. This demonstrates that the web page hosted at the server address was reachable from the test PC.



Screenshot 11 — Ping tests from a client

This command-prompt screenshot records ping tests to addresses in the IT, Sales and HR ranges. It includes both successful and unsuccessful attempts. Ping is a reachability test; a failure can have several causes and should be interpreted alongside device and router configuration.



Screenshot 12 — HR reachability and Sales-to-IT ping result

The ping to 192.168.30.5 succeeds with four packets received and 0% packet loss. The ping from Sales to 192.168.10.5 fails with “Destination host unreachable” reported by 192.168.20.1. This supports the observed test results, but the ping output alone does not prove which ACL rule caused the restriction.

